Experiment-2 (smart traffic controller)

**Aim:**

To design a traffic signal controller that adjusts green light duration based on traffic density.

**Algorithm:**

1. Accept traffic density input from the user (`low`, `medium`, or `high`).
2. If traffic density is `low`, set green signal to 10 seconds.
3. If traffic density is `medium`, set green signal to 20 seconds.
4. If traffic density is `high`, set green signal to 30 seconds.
5. If input is invalid, display an error message.
6. Print confirmation that the traffic signal controller is activated.

**Python Program:**

```
python
traffic = input("Enter traffic density (low/medium/high): ")

if traffic == "low":
    print("Green signal: 10 seconds")
elif traffic == "medium":
    print("Green signal: 20 seconds")
elif traffic == "high":
    print("Green signal: 30 seconds")
else:
    print("Invalid traffic density")

print("Traffic signal controller activated")
```

**Sample Output 1 (for low traffic):**

```
Enter traffic density (low/medium/high): low
Green signal: 10 seconds
Traffic signal controller activated
```

**Sample Output 2 (for high traffic):**

Experiment-2 (smart traffic controller)

```
Enter traffic density (low/medium/high): high
Green signal: 30 seconds
Traffic signal controller activated
```